HAND & UPPER LIMB · NEET-SS / FET / INI-SS

Brachial plexus

Anatomy

- Roots C5-T1 (prefixed from C4, postfixed to T2), trunks (upper C5-6, middle C7, lower C8-T1), divisions (behind the clavicle), cords (lateral, posterior, medial, named by position around the axillary artery), terminal branches.
- **Branches from the roots:** dorsal scapular (C5, rhomboids), **long thoracic (C5-7, serratus anterior)**, and the phrenic nerve contribution (C3-5).
- **Upper trunk branches:** suprascapular, nerve to subclavius.

Preganglionic (root avulsion) vs postganglionic injury

- **Signs of a preganglionic lesion:**
  - **Horner syndrome** (T1 sympathetic fibres).
  - **Winged scapula** (long thoracic) and rhomboid paralysis (dorsal scapular).
  - Raised hemidiaphragm (phrenic).
  - Severe burning **neuropathic pain**.
  - **No Tinel sign** in the neck.
  - **Pseudomeningocele** on CT myelography or MRI.
  - **Preserved SNAPs in an anaesthetic limb**: the dorsal root ganglion cell body is still attached to its peripheral axon. This is the classic electrodiagnostic sign.
- Preganglionic injuries **cannot be repaired**; function is restored with **nerve transfers**. Postganglionic ruptures can be grafted.

Patterns

- **Erb (C5-6):** "waiter's tip": adducted, internally rotated shoulder, extended elbow, pronated forearm.
- **Klumpke (C8-T1):** clawing and intrinsic paralysis, often with Horner syndrome.
- **Total plexus:** flail, anaesthetic arm.

Adult traction injuries (motorcycle accidents)

- **Timing:** sharp open injuries need immediate exploration and repair. Closed injuries: observe for about **3 months** for signs of recovery, but reconstruct **before about 6 months**, so reinnervation beats motor end-plate loss.
- **Priorities: elbow flexion** first, then **shoulder abduction and external rotation and stability**, then hand function.
- **Key nerve transfers:**

Goal	Transfer
Elbow flexion	**Oberlin:** an ulnar nerve fascicle (to FCU) to the biceps branch of the musculocutaneous nerve. **Double fascicular (Mackinnon):** add a median fascicle to the brachialis branch
Shoulder abduction / external rotation	**Spinal accessory to suprascapular** nerve
Deltoid	**Long head of triceps branch of the radial nerve to the axillary nerve**
Total plexus	Intercostals or contralateral C7 to musculocutaneous and other targets

- **Late presentation (over 12-18 months):** tendon transfers, **free functioning gracilis** muscle transfer, shoulder arthrodesis.

Obstetric brachial plexus palsy

- Risk factors: macrosomia, **shoulder dystocia**, assisted delivery. Most are Erb type and most recover.
- **Return of biceps function by 3 months** predicts good recovery (Gilbert-Tassin). If biceps has not recovered by **3-6 months**, explore with grafting or nerve transfers. Narakas grades severity.
- Sequelae: **internal rotation contracture** leading to posterior subluxation and **glenohumeral dysplasia**; assessed with the **Mallet score**.
- Treatment: release of subscapularis, and **latissimus dorsi and teres major transfer** to the rotator cuff for external rotation (L'Episcopo-type procedures). Humeral derotation osteotomy in older children.
- Differentials: clavicle or humerus fracture (pseudoparalysis), septic shoulder.

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.